Global Illumination Methods

Practical Course

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graphics.uni.kn

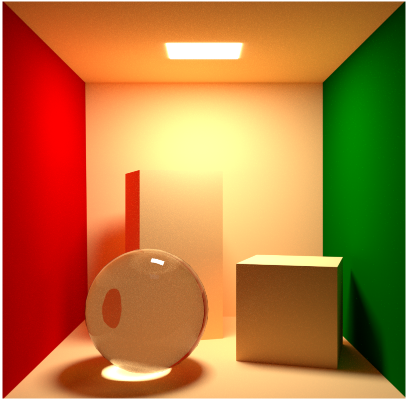
Organizational and Scope

Course Goal

Students are expected to implement their own ray tracer during the practical course.

What this means

- start from a runnable framework,
- extend it step by step during the semester,
- explain your implementation decisions in the final oral examination.



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Exercise Format

- C++ template provided, working as a starter renderer with a few code fill-in tasks.
- 4 project themes roughly defined - autonomous exploration required!
- Work alone or in teams of two.
- Use of AI tools is encouraged.
- Individual oral examination, including course contents, code review and discussion of your own implementation.

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Template Download

GitLab:
https://gitlab.inf.uni-konstanz.de/visual-computing/lecture-global-illumination-methods/lecture-global-illumination-methods-2026

Alternative:
download the project as a zip file from the course website.

Course website:
https://www.cgmi.uni-konstanz.de/en/student-corner/summer-semester-26/global-illumination-methods/

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Project Themes

1. Geometry, Scene Representation & Acceleration
2. Appearance, Materials & Light Interaction
3. Light Transport & Global Illumination
4. Camera Optics, Sampling & Participating Media

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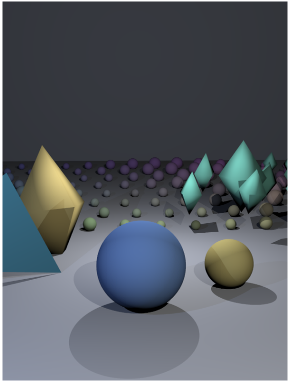
Theme 1: Geometry, Scene Representation & Acceleration

How should a scene be represented, organized, and queried so that rays can find visible surfaces correctly and efficiently?

- analytic intersections, triangle meshes, hit records, normals
- scene hierarchies, bounding boxes, BVH or other acceleration structures
- ideal for students who enjoy geometry, algorithms, and performance



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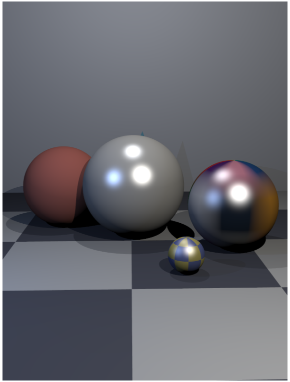
Theme 2: Appearance, Materials & Light Interaction

Once a ray hits a surface, how do materials, textures, light sources, and visibility determine the visible appearance?

- Lambertian and local shading models, BRDF basics
- texture mapping, UV coordinates, point and directional lights
- a strong fit for visual appearance, shading behavior, and lighting



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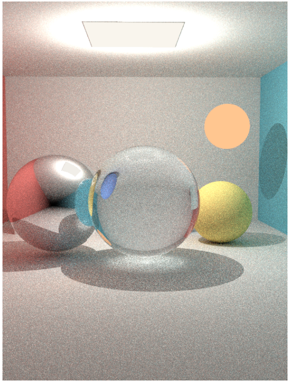
Theme 3: Light Transport & Global Illumination

How can the renderer go beyond direct illumination and account for reflection, refraction, and indirect light transport?

- recursive ray tracing, mirror reflection, refraction, Fresnel effects
- Whitted-style rendering, path tracing, Monte Carlo integration
- best for students interested in the core light transport problem



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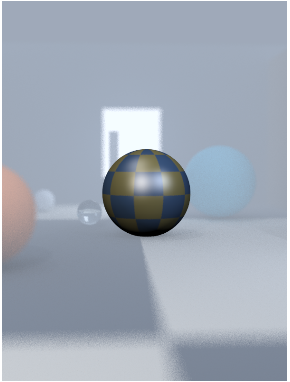
Theme 4: Camera Optics, Sampling & Participating Media

How do sampling strategies, camera models, and volumetric effects influence the final image beyond the simplest pinhole camera?

- anti-aliasing, multi-sampling, stratified or jittered sampling
- thin lens camera, depth of field, fog, transmittance, media effects
- a good fit if you enjoy image quality, camera models, and fog



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